SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Answer Script Evaluation Sheet

Assignment 1

Date:05-11-26

Programme	B.E/ B. Tech	**Slot**	C
Course Code	CSA0709	**Course Name**	Computer Networks
Duration	1.30 Hrs	**Marks**	50

Q. No.	Max. Marks	Awarded Marks
Total	50	

Course Faculty: Dr. Shankar R.,

TASK 1 – Size the Traffic Before Choosing Anything

1. Calculation of Network Traffic

Given Data:

Source	**Quantity**	**Traffic**
Robot controllers	40	200 bytes/2 ms
Vibration sensors	600	50 bytes/s
CCTV cameras	24	4 Mbps each
Office users	150	2 Mbps each (10% active)

A. Robot Controllers

$$\frac{200\times8}{0.002\times10^{6}} = 0.8Mbps$$

40 times 0.8=32 Mbps

B. Vibration Sensors

600 times50 * times * 8 / 10^6 = 0.24 Mbps

C. CCTV Cameras

24 times * 4= 96 Mbps

D. Office Users

Active users = 150 × 10% = 15

15 * times2= 30 Mbps

2. Total Network Traffic

One production line:

32+0.24+96+30= 158.24 Mbps

Entire plant (8 lines):

158.24 times * 8= 1265.92 Mbps

3. Highest Traffic and Strictest Delay

- Highest traffic: CCTV cameras (96 Mbps).
- Strictest delay: Robot controllers (10 ms requirement).
- They are not the same because CCTV generates more data, while robots require faster communication.

4. Is a 1 Gbps Uplink Sufficient?

Required bandwidth = 1265.92 Mbps.

Available bandwidth = 1000 Mbps.

$$1265.92 - 1000 = 265.92 Mbps$$

Therefore, a 1 Gbps uplink is insufficient.

If CCTV and robot packets share the same FIFO queue, robot commands may experience delays and packet loss.

Solution: Use QoS to give robot commands the highest priority and keep CCTV traffic in a separate queue.

5. Ranking Based on 100 ms Delay

Rank	Traffic	Impact
1	Robot controllers	Violates the 10 ms requirement and affects machine operations.
2	CCTV cameras	Delays live video monitoring.
3	Vibration sensors	Delays equipment fault detection.
4	Office users	Slows ERP and email access.

6. Alternative Concept – Local CCTV Recording

If CCTV cameras record locally and transmit clips only 5% of the time:

Original CCTV traffic = 96 Mbps.

$96 \times \frac{5}{100} = 4.8 Mbps$

New traffic per line:

32+0.24+4.8+30 = 67.04 Mbps

New traffic for 8 lines:

67.04 times * 8= 536.32 Mbps

Uplink utilization:

536.32 * 1000 / 100= 53.63

1 Gbps is sufficient on average

7. Comparison

Parameter	**Continuous streaming**	**Local recording**
CCTV traffic/line	96 Mbps	4.8 Mbps
Plant traffic	1265.92 Mbps	536.32 Mbps
1 Gbps uplink	Insufficient	Sufficient on average
Congestion	High	Reduced

8. Conclusion

Continuous CCTV streaming increases network traffic and makes the 1 Gbps uplink insufficient. Local CCTV recording reduces bandwidth consumption and makes the existing uplink sufficient under average conditions. QoS should prioritize robot commands to maintain the 10 ms delay requirement. Therefore, local recording, traffic prioritization and scalable network infrastructure are recommended for reliable plant communication.

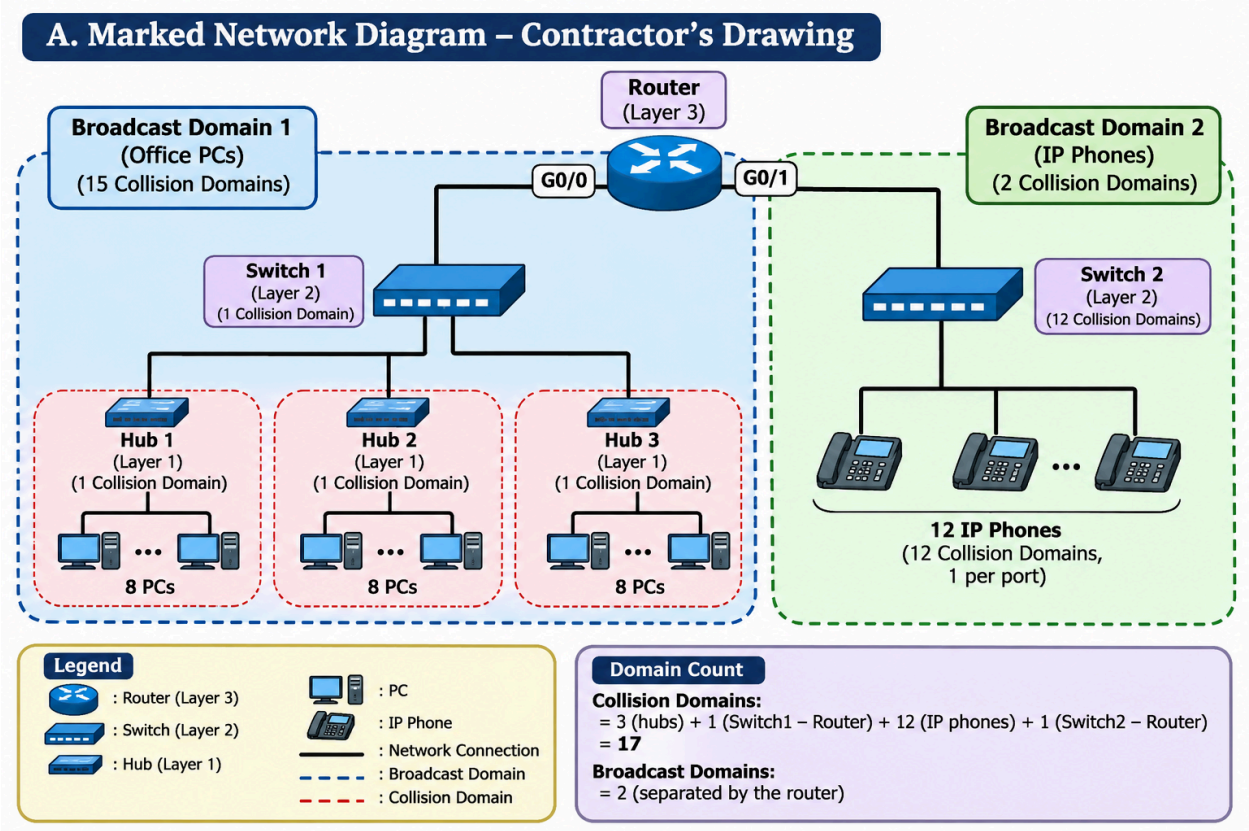
TASK 2 – Count the Domains in the Contractor's Drawing

1. Collision Domains and Broadcast Domains

Given network:

- Three hubs with 8 PCs each.
- All hubs connect to one 24-port switch.
- Switch connects to router interface G0/0.
- Router interface G0/1 connects to another switch with 12 IP phones.

A. Marked Network Diagram



B. Number of Domains

Collision domains:

- Each hub creates one shared collision domain.
- Three hubs = 3 collision domains.
- Switch 1 to router G0/0 = 1 collision domain.
- Switch 2 has 12 phone ports and one router connection = 13 collision domains.

3+1+13= 17

Broadcast domains:

A router separates broadcast domains.

2 Broadcast Domains

2. OSI Layers of Network Devices

Device	**OSI Layer**	**Function**
Hub	Layer 1 – Physical	Transfers electrical signals.
Switch	Layer 2 – Data Link	Forwards frames using MAC addresses.
Router	Layer 3 – Network	Routes packets using IP addresses.
PC	All 7 layers	Sends and receives network data.
IP Phone	Multiple layers	Supports voice communication using network protocols.

Why can't a hub stop collisions?

A hub works at the Physical Layer and sends incoming signals to all connected devices. It does not examine MAC addresses or separate traffic. Therefore, all connected devices share the same collision domain.

3. Types of Connections

Connection	**Type**
PCs connected to hubs	Multipoint
Hubs connected to switch	Multipoint shared Ethernet segment
Switch connected to router	Point-to-point
IP phones connected to switch	Point-to-point

4. Network Standards and Organizations

Technology	**Standard**	**Organization**
Ethernet	IEEE 802.3	IEEE
Wi-Fi	IEEE 802.11	IEEE

IPv4	RFC 791	IETF
OSI Model	ISO 7498	ISO
5G	IMT-2020 / 3GPP specifications	ITU / 3GPP

Why should the plant use standard-compliant equipment?

Standard-compliant equipment ensures compatibility between devices from different manufacturers. It improves interoperability, reliability, security and network scalability while reducing maintenance costs.

5. Alternative Concept Analysis – Replacing Hubs with Switches

If the three hubs are replaced with three unmanaged switches:

Collision domains:

Each access switch has 8 PC connections and 1 uplink connection.

$$3 \times (8 + 1) = 27$$

Adding the router connection on Switch 1:

$$27+1+13= 41$$

Broadcast domains:

2

Comparison

Parameter	Original (Hubs)	Replaced (Switches)
Collision domains	17	41
Broadcast domains	2	2
Collisions	More likely	Greatly reduced
Performance	Lower	Better
Cost	Lower	Higher

What does this change fix?

- Reduces collisions through separate switch ports.
- Improves network performance.
- Provides dedicated bandwidth per switched link.

What does it not fix?

Unmanaged switches do not separate broadcast domains. Broadcast traffic can still reach all devices within the same VLAN.

6. Conclusion

The contractor's original network contains 17 collision domains and 2 broadcast domains. Replacing the hubs with switches increases collision domains to 41 and reduces collisions, improving network performance. However, broadcast domains remain unchanged. Therefore,

switches are a better choice than hubs for the smart manufacturing plant because they provide better efficiency, scalability and reliable communication.

TASK 3 – Read a Captured Frame Layer by Layer

1. OSI Layers and Protocol Data Units (PDU)

The captured robot command passes through different OSI layers. Each layer adds its own information to ensure correct delivery.

Captured field	**OSI Layer**	**PDU**
Ethernet II	Layer 2 – Data Link	Frame
IPv4	Layer 3 – Network	Packet
TCP	Layer 4 – Transport	Segment
Data	Layers 5–7	Data

2. Number of Routers Crossed

Given:

- Initial TTL = 64
- Received TTL = 61

64 - 61 = 3 routers

Therefore, the packet crossed 3 routers. Each router decreases the TTL value by one.

3. Total Bytes and Overhead Calculation

A. Robot Command (12 bytes)

Given:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Data = 12 bytes
- FCS = 4 bytes

Total frame size:

$$14+20+20+12+4 = 70 \text{ bytes}$$

Overhead:

$$14 + 20 + 20 + 4 = 58 bytes$$

Overhead percentage:

$$58 / 70 * 100 = 82.86\ \%$$

B. Camera Packet (1,400 bytes)

$$14+20+20+1400+4 = 1458 \text{ bytes}$$

Overhead percentage:

$$58 \setminus 1458 * 100 = 3.98\ \%$$

Evaluation: Small robot commands have a higher overhead percentage because protocol headers consume a large portion of the transmitted data. Large camera packets use bandwidth more efficiently.

4. Fault-to-OSI Layer Mapping

Problem	**Responsible layer**	**Reason**
Wrong MAC address	Layer 2 – Data Link	MAC addresses identify devices on the local network.
TTL reaches 0	Layer 3 – Network	IP routers discard packets when TTL becomes zero.
Destination port closed	Layer 4 – Transport	TCP checks destination ports and may return a reset (RST).

5. TCP/IP Model Mapping

TCP/IP Model	**Corresponding OSI Layers**
Application	Layers 7, 6 and 5
Transport	Layer 4
Internet	Layer 3

Network Access	Layers 2 and 1

The TCP/IP model combines some OSI layers but performs similar communication functions.

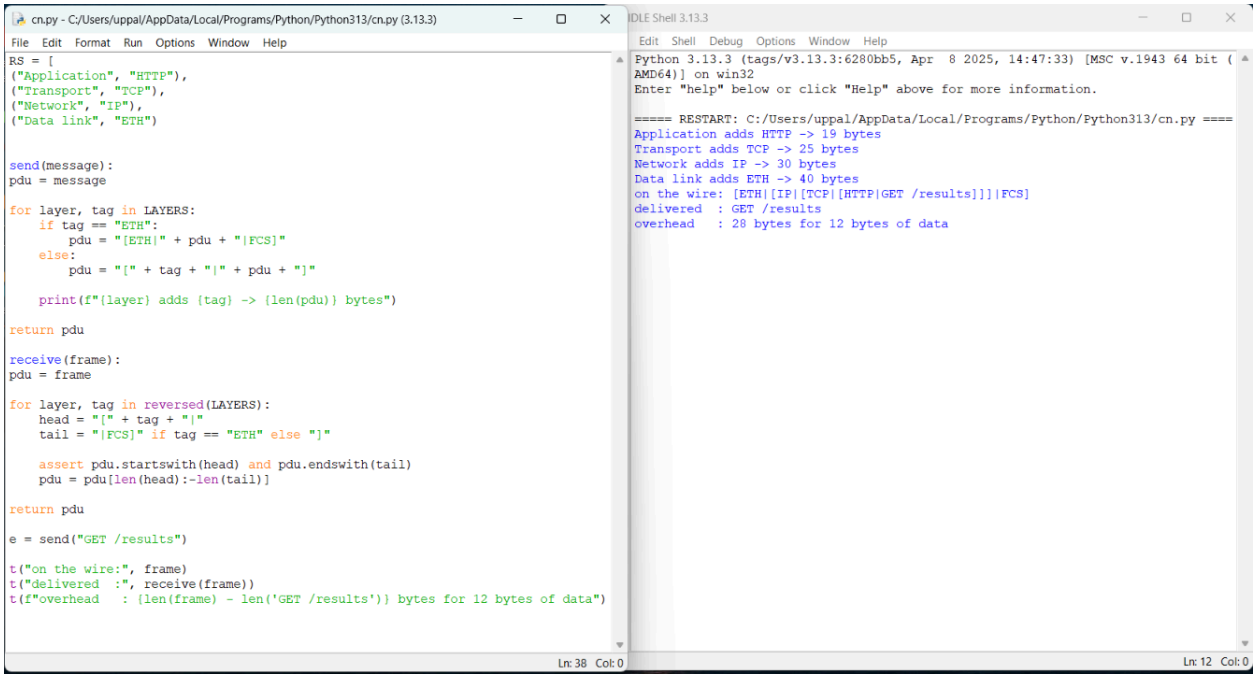
6. Complete the Code – Python Program

Program:

```
LAYERS = [
    ("Application", "HTTP"),
    ("Transport", "TCP"),
    ("Network", "IP"),
    ("Data link", "ETH")
]
def send(message):
    pdu = message
    for layer, tag in LAYERS:
        if tag == "ETH":
            pdu = "[ETH|" + pdu + "|FCS]"
        else:
            pdu = "[" + tag + "|" + pdu + "]"
        print(f"{layer} adds {tag} -> {len(pdu)} bytes")
```

```
    return pdu
def receive(frame):
    pdu = frame
    for layer, tag in reversed(LAYERS):
        head = "[" + tag + "|"
        tail = "|FCS]" if tag == "ETH" else "]"
        assert pdu.startswith(head) and pdu.endswith(tail)
        pdu = pdu[len(head):-len(tail)]
    return pdu
frame = send("GET /results")
print("on the wire:", frame)
print("delivered  :", receive(frame))
print(f"overhead   : {len(frame) - len('GET /results')} bytes for 12 bytes of data")
```

Output



Change the Message to 1,400 Bytes

frame = send("A" * 1400)

Synthetic program overhead remains 28 bytes.

$$28 / 1400+28 * 100 = 1.96\ \%$$

Therefore, larger messages have a lower overhead percentage.

7. Alternative Concept Analysis – Direct Ethernet

An industrial protocol can transmit robot commands directly using Ethernet with EtherType 0x8892, without IP and TCP headers.

Bytes saved per command:

$$20+20 = 40 \text{ bytes}$$

Bandwidth saved per command:

$$40 \times 8 = 320bits$$

Commands per second equivalent to saved bandwidth:

$$100{,}000{,}000 / 320 = 312{,}500$$

This means the saved header bandwidth is equivalent to 312,500 command-sized overhead savings per second on a 100 Mbps link.

Advantages:

- Reduces protocol overhead.
- Improves bandwidth efficiency.
- Supports fast industrial communication.

Disadvantages:

- Loses normal IP routing.
- Requires specialized industrial network equipment.
- Reduces compatibility with ordinary IP-based networks.

8. Conclusion

The captured frame shows how Ethernet, IPv4 and TCP work together to deliver robot commands. The packet crossed three routers, and small commands experienced a high overhead percentage compared with large camera packets. Direct Ethernet communication can reduce overhead, but IP provides better routing and compatibility. Therefore, the plant should select protocols based on latency, reliability and network requirements.

TASK 4 – Choose a Medium for Each Run (There Is a Trap)

1. Selection of Transmission Media

Cable run	Distance	Recommended medium	Reason
Robot cell near welding machines	85 m	Cat6A S/FTP	Shielding protects against electromagnetic interference.
Office to shop-floor switch	140 m	Fibre optic	Exceeds the 100 m copper Ethernet limit.
Shop floor to data centre	550 m	Multimode fibre	Suitable for high-speed connections over this distance with compatible optics.
Plant to head office	400 km	Single-mode fibre / leased line	Supports long-distance, high-speed communication.

2. Which Run Cannot Use Copper Ethernet?

Standard twisted-pair Ethernet supports a maximum channel length of approximately 100 metres.

Therefore:

- 85 m – Copper Ethernet is suitable.
- 140 m – Copper Ethernet is unsuitable.
- 550 m – Copper Ethernet is unsuitable.
- 400 km – Copper Ethernet is unsuitable.

Trap: The 140 m connection cannot use ordinary copper Ethernet even though it exceeds the limit by only 40 metres. Fibre optic cable is recommended.

3. Shannon Capacity Calculation

Shannon's formula:

$$C = B\log_2(1 + SNR)$$

Where:

- C = Channel capacity
- B = Bandwidth in Hz
- SNR = Signal-to-noise ratio

A. Cable near normal conditions (40 dB)

Given:

- Bandwidth = 250 MHz
- SNR = 40 dB

Convert SNR:

$$SNR = 10^{40/10} = 10000$$

Apply Shannon formula:

$$C = 250 \times 10^6 \log_2(10001)$$

C = 3321.96 Mbps

B. Cable near welding machines (15 dB)

$$SNR = 10^{15/10} = 31.62$$

$$C = 250 \times 10^{6} \log_2(1 + 31.62)$$

C = 1256.95 Mbps

Evaluation: Welding machines increase electrical interference and reduce SNR. The theoretical capacity decreases from approximately 3321.96 Mbps to 1256.95 Mbps. Therefore, shielded cables are recommended for reliable robot communication.

4. Why Actual Speed Is Below Shannon Capacity

Actual network speed is usually lower than the Shannon limit because of:

1. Electrical and electromagnetic interference.
2. Protocol overhead.
3. Hardware limitations.
4. Signal attenuation.
5. Retransmissions and packet loss.

Therefore, Shannon capacity represents a theoretical maximum, not the actual transmission speed.

5. Complete the Code – Python Program

```
import math

def nyquist(bandwidth_hz, levels):
    return 2 * bandwidth_hz * math.log2(levels)

def shannon(bandwidth_hz, snr_db):
    snr = 10 ** (snr_db / 10)
```

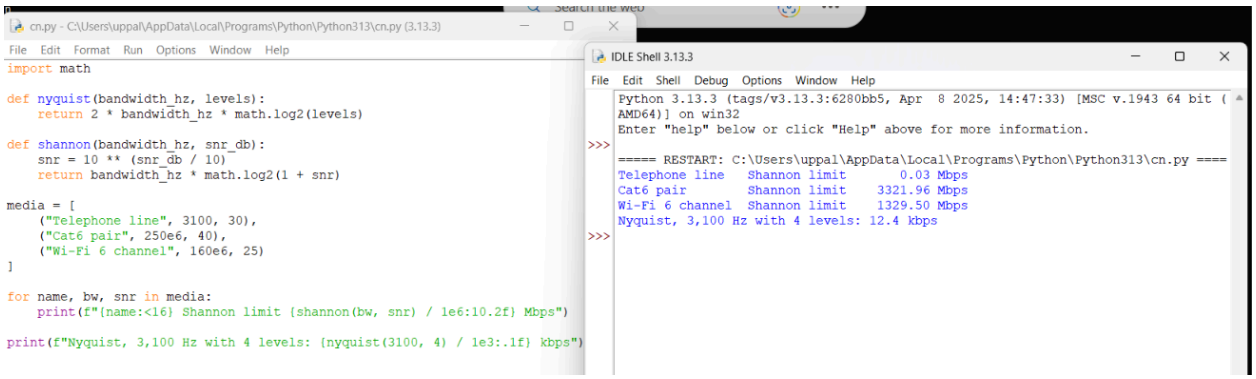
```
    return bandwidth_hz * math.log2(1 + snr)

media = [
    ("Telephone line", 3100, 30),
    ("Cat6 pair", 250e6, 40),
    ("Wi-Fi 6 channel", 160e6, 25)
]

for name, bw, snr in media:
    print(f"{name:<16} Shannon limit {shannon(bw, snr) / 1e6:10.2f} Mbps")

print(f"Nyquist, 3,100 Hz with 4 levels: {nyquist(3100, 4) / 1e3:.1f} kbps")
```

Output



6. Telephone Line SNR for 56 kbps

Given:

- Bandwidth = 3100 Hz
- Required capacity = 56 kbps

Using Shannon formula:

$$56000 = 3100\log_2(1 + SNR)$$

$$SNR = 2^{56000/3100} - 1$$

SNR ~~ 54.38 dB

Evaluation: This is a very high SNR requirement for a telephone line. Therefore, achieving 56 kbps over this bandwidth is difficult under practical conditions.

7. Alternative Concept Analysis – Wi-Fi 6 Near Welders

Given:

- Bandwidth = 160 MHz
- SNR = 15 dB

$$SNR = 10^{15/10} = 31.62$$

$$C = 160 \times 10^6 \log_2(32.62)$$

C = 804.45 Mbps

Evaluation:

Although Wi-Fi 6 provides high theoretical capacity, welding interference, wireless contention, jitter and variable latency can affect robot commands.

Therefore, Wi-Fi 6 should not be the primary connection for critical robot control requiring 10 ms latency. Shielded Ethernet or industrial fibre with appropriate real-time networking is preferred.

8. Conclusion

Different transmission media are selected according to distance, interference and bandwidth requirements. Fibre optic cables are suitable for long-distance communication, while shielded Cat6A is preferred near welding machines. Shannon's theorem shows that interference reduces theoretical channel capacity. Therefore, wired and shielded communication is recommended for reliable industrial operations.

TASK 5 – Wi-Fi 6 or 5G for Moving Equipment?

1. Hand-over Calculation

Given:

- AGV speed = 2 m/s
- Route distance = 1.2 km = 1200 m
- Wi-Fi AP spacing = 30 m
- Wi-Fi hand-over time = 150 ms
- 5G cell coverage = 300 m
- 5G hand-over time = 20 ms

A. Wi-Fi 6

Number of coverage sections:

$$1200/30 = 40$$

Hand-overs:

$$40 - 1 = 39$$

Total unreachable time:

$$39 \times 150 = 5850ms$$

5.85 seconds

B. Private 5G

Number of coverage sections:

$$1200/300 = 4$$

Hand-overs:

$$4 - 1 = 3$$

Total unreachable time:

$$3 \times 20 = 60ms$$

0.06 seconds

Comparison

Parameter	**Wi-Fi 6**	**Private 5G**
Coverage sections	40	4
Hand-overs	39	3
Handover time	150 ms	20 ms

Total unreachable time	5850 ms	60 ms

Evaluation: Private 5G provides significantly fewer hand-overs and lower communication downtime. However, even its 20 ms individual hand-over exceeds the AGV's 10 ms command deadline.

2. 5G Service Types

5G service	**Application**	**Reason**
URLLC	AGVs and robots	Requires ultra-low latency and high reliability.
eMBB	CCTV cameras	Supports high-bandwidth video streaming.
mMTC	Temperature and vibration sensors	Supports large numbers of connected devices.

3. Uses of Bluetooth, Zigbee and LoRa

Technology	**Suitable application**
Bluetooth	Short-range wireless devices and maintenance equipment.

Zigbee	Low-power sensor networks.
LoRa	Long-distance, low-data-rate monitoring sensors.

These technologies are useful when Wi-Fi or 5G is unnecessary or too expensive.

4. Can Wi-Fi 6 with Fast Roaming (802.11r) Meet the Deadline?

IEEE 802.11r enables fast roaming between Wi-Fi access points by reducing authentication delays.

Advantages:

- Faster access-point transitions.
- Reduced hand-over delay.
- Improved wireless mobility.

Limitations:

- It does not guarantee a delay below 10 ms.
- Interference can affect communication.
- Packet loss and jitter may occur.
- Actual performance depends on network configuration.

Decision: Wi-Fi 6 with 802.11r may improve roaming, but it should not be assumed to meet the strict 10 ms requirement without testing.

5. Alternative Concept Analysis – LoRaWAN for Sensors

Given:

- Number of sensors = 600
- Data per sensor = 50 bytes/second

- LoRa gateway capacity = 5 kbps

Step 1: Calculate total sensor data.

$$600 \times 50 = 30000 bytes/s$$

Step 2: Convert into bits.

$$30000 \times 8 = 240000 bits/s$$
$$= 240 \text{ kbps}$$

Step 3: Compare with gateway capacity.

$$240/5 = 48$$

The required bandwidth is 48 times greater than the assumed gateway capacity.

LoRaWAN capacity is insufficient

Evaluation:

- LoRaWAN is suitable for low-frequency, small sensor messages.
- All 600 sensors cannot transmit at the given rate through one 5 kbps gateway.
- Sensors requiring continuous vibration monitoring should remain on Wi-Fi or wired Ethernet.
- Low-frequency temperature and environmental sensors can use LoRaWAN if their reporting rates are reduced.

6. Conclusion

Private 5G is a better choice than ordinary Wi-Fi for moving AGVs because it provides fewer hand-overs and lower downtime. URLLC is suitable for critical robot communication, eMBB for CCTV and mMTC for sensors. LoRaWAN can support low-data-rate monitoring but cannot

handle all 600 sensors at the given data rate. Therefore, a combination of private 5G, wired Ethernet, Wi-Fi and LoRaWAN is recommended for the smart manufacturing plant.

TASK 6 – Interpret an Ethernet Port Report

1. Interpretation of Ethernet Port Report

Given data:

Parameter	Reported value
Speed	100 Mbps
Duplex	Half
Auto-negotiation	Off
Link detected	Yes
RX errors	2381
Late collisions	412
CRC errors	1907

A. Meaning of Late Collision and CRC Error

Late Collision:

A late collision occurs when two devices transmit simultaneously and a collision is detected after the normal Ethernet collision window. It can indicate duplex mismatch or cabling problems.

CRC Error:

A CRC error occurs when the received frame fails its Cyclic Redundancy Check. It indicates that data may have been corrupted during transmission due to interference, damaged cables or duplex mismatch.

Evaluation: The combination of late collisions and CRC errors strongly suggests a duplex mismatch or a physical cabling problem.

2. Likely Cause and Configuration Correction

The likely cause is that the shop-floor PC is manually configured for 100 Mbps half-duplex, while the gigabit switch port is configured for auto-negotiation.

This mismatch can cause collisions, retransmissions and slow network performance.

Linux Configuration Using ethtool

First, check the network interface:

sudo ethtool eth0

Enable auto-negotiation:

sudo ethtool -s eth0 autoneg on

Check the configuration again:

sudo ethtool eth0

Both ends should use compatible speed and duplex settings. Auto-negotiation on both ends is recommended.

Windows Configuration:

1. Open Device Manager.
2. Select Network Adapters.
3. Right-click the Ethernet adapter.
4. Open Properties and select Advanced.
5. Select Speed & Duplex.
6. Choose Auto Negotiation.
7. Click OK.

3. MAC Address Analysis

A. MAC Address: 01:00:5e:0a:01:02

This is a multicast MAC address, not the address of a single device.

It is used to deliver Ethernet frames to a group of devices, particularly for IPv4 multicast communication.

B. MAC Address: 3c:52:82:4e:11:07

The first three bytes are:

3C:52:82

These bytes represent the OUI (Organizationally Unique Identifier), which identifies the manufacturer or organization assigned that MAC address range.

The remaining three bytes identify a particular interface within that range.

4. Jumbo Frames – 9,000 Bytes

Jumbo frames are Ethernet frames that carry larger payloads than standard Ethernet frames (typically 1,500-byte MTU).

Would jumbo frames help the CCTV storage server?

Yes. They can:

- Reduce the number of frames transmitted.
- Reduce protocol processing overhead.
- Improve efficiency during large video transfers.
- Reduce CPU processing load in some situations.

Requirements:

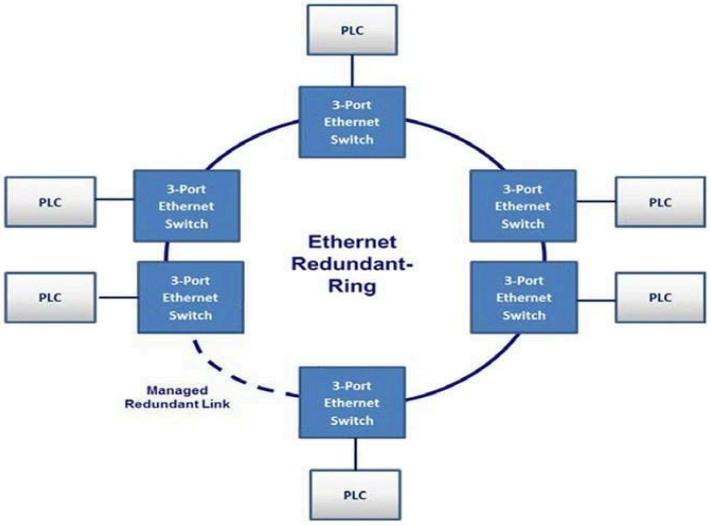
All devices along the communication path must support and be configured for the same jumbo-frame MTU, including:

- CCTV storage server NIC.
- Network switches.
- Camera interfaces, if applicable.
- Router interfaces, if traffic passes through routers.

Jumbo frames will not automatically improve CCTV performance if the network is limited by storage speed or camera bandwidth.

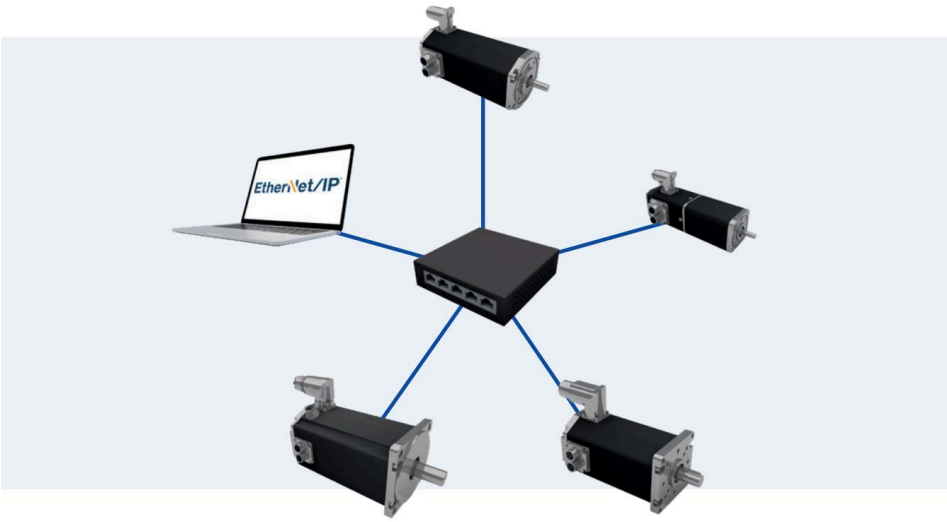
5. Alternative Concept Analysis – Ring vs Star Topology

A. Ring Topology



- Industrial switches are connected in a closed loop.
- If one cable is cut, traffic can use the alternative path after ring recovery.
- Fast ring recovery protocols can restore communication in less than 50 ms, depending on the technology and configuration.
- Devices generally remain connected after recovery.

B. Star Topology



Graphic: Schematic representation - Star Topology

- All devices connect to a central switch.
- If one device cable is cut, only that device loses connection.
- Other devices continue communicating.
- If the central switch or its uplink fails, multiple connected devices may lose communication.

Comparison

Parameter	**Ring**	**Star**
Topology	Circular	Central switch

Cable cut	Alternative path available	Affected connection fails
Recovery	Fast ring recovery	Usually no recovery needed for other links
Reliability	High with redundancy	Depends on central switch
Cost	Comparatively higher	Generally lower

Evaluation: Ring topology is suitable for industrial environments requiring redundant communication, while star topology is simpler and easier to maintain.

6. Conclusion

The Ethernet report indicates possible duplex mismatch and transmission errors, which can reduce network performance. Enabling auto-negotiation on both ends can help resolve the mismatch. Multicast MAC addresses identify groups of devices, while OUI identifies the assigned manufacturer. Jumbo frames can improve large CCTV data transfers when all network devices support them. Therefore, a properly configured Ethernet network with redundant ring connections is recommended for reliable smart manufacturing communication.

TASK 7 – Will the Receiver Notice the Error?

1. CRC Calculation Using Modulo-2 Division

Given:

- Data = 1101011011
- Generator = 1011
- Generator polynomial = $x^3 + x + 1$

Since the generator has 4 bits, append 3 zeros to the data.

Data : 1101011011
Append 000 : 1101011011000
Divisor : 1011

Perform modulo-2 division using XOR (exclusive OR).

1101011011000 ÷ 1011

The remainder obtained is:

100

Therefore:

- CRC bits = 100
- Transmitted frame = 1101011011100

2. Check Whether the Received Frame Is Accepted

Given received frame:

1101011011101

Divide it by generator 1011.

Remainder:

001

Since the remainder is not zero, the receiver rejects the frame.

Result:

Frame Rejected

3. Burst Error Detection

The generator 1011 has degree 3.

- It can detect all burst errors of length 3 bits or less.
- Longer burst errors may or may not be detected.
- An error pattern divisible by the generator cannot be detected.

Example of an undetected error:

1011000000000

This pattern is the generator 1011 shifted by zeros. Its modulo-2 division gives remainder 000.

Therefore, the CRC cannot detect this error.

4. Two-Dimensional Parity

Consider the following 4 × 4 data block using even parity.

1 0 1 1 | 1
0 1 0 1 | 0
1 1 0 0 | 0
0 0 1 0 | 1
---
0 0 0 0

The last column contains row parity bits, and the last row contains column parity bits.

Advantages:

- Detects single-bit errors.
- Identifies the row and column containing a single-bit error.
- Improves error detection compared with simple parity.

Undetected error pattern:

If four corner bits of a rectangular block are changed, each affected row and column has two errors. Even parity remains unchanged, so the error is not detected.

5. Internet Checksum Calculation

Given four 16-bit hexadecimal words:

4500
0073
0000
4000

Add all words:

$$4500 + 0073 + 0000 + 4000 = 8573$$

Take the one's complement:

$$FFFF - 8573$$
$$7A8C$$

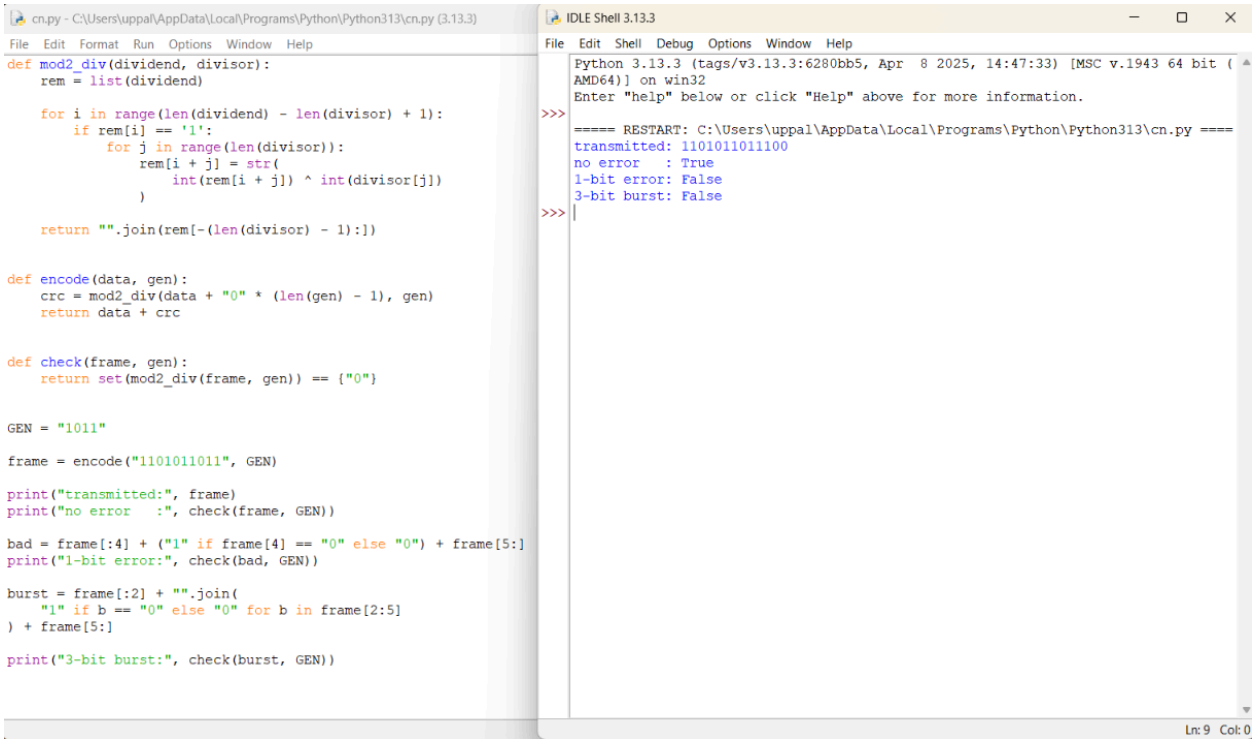
Therefore, the Internet checksum is 7A8C (hexadecimal).

6. Complete the Code – Python Program

```
def mod2_div(dividend, divisor):
    rem = list(dividend)
    for i in range(len(dividend) - len(divisor) + 1):
        if rem[i] == '1':
            for j in range(len(divisor)):
```

```
            rem[i + j] = str(
                int(rem[i + j]) ^ int(divisor[j])
            )
    return "".join(rem[-(len(divisor) - 1):])
def encode(data, gen):
    crc = mod2_div(data + "0" * (len(gen) - 1), gen)
    return data + crc
def check(frame, gen):
    return set(mod2_div(frame, gen)) == {"0"}
GEN = "1011"
frame = encode("1101011011", GEN)
print("transmitted:", frame)
print("no error   :", check(frame, GEN))
bad = frame[:4] + ("1" if frame[4] == "0" else "0") + frame[5:]
print("1-bit error:", check(bad, GEN))
burst = frame[:2] + "".join(
    "1" if b == "0" else "0" for b in frame[2:5]
) + frame[5:]
print("3-bit burst:", check(burst, GEN))
```

Output



7. Alternative Concept Analysis – CRC vs Even Parity

Even parity adds one extra bit to make the total number of 1s even.

Error pattern	Even parity	CRC-3
2-bit error 0000000000011	Not detected	Detected
4-bit error 0000000011110	Not detected	Detected
4-bit error 1011000000000	Not detected	Not detected

Evaluation:

- Even parity detects odd numbers of bit errors but misses even numbers of errors.
- CRC detects all single-bit errors and burst errors up to 3 bits.
- CRC may miss certain longer error patterns that are divisible by its generator polynomial.

Therefore, CRC provides stronger error detection than simple even parity.

8. Conclusion

CRC is an effective error-detection technique used in Ethernet communication. The generator 1011 detects single-bit errors and burst errors up to 3 bits. Two-dimensional parity provides additional error detection, while the Internet checksum verifies data integrity. Therefore, CRC is recommended for reliable communication between industrial network switches.

TASK 8 – Correct the Error or Ask Again?

1. Hamming Check Bits Calculation

Given:

- Data bits (m) = 16
- Check bits = r

Formula:

$$2^r \geq m + r + 1$$

Try $r = 4$:

$$16 < 16 + 4 + 1 = 21$$

Not sufficient.

Try $r = 5$:

$$32 \geq 16 + 5 + 1 = 22$$

Therefore:

$$r = 5 \text{ check bits}$$

Total bits:

$$16 + 5 = 21 bits$$

Overhead percentage:

$$5 / 21 * 100 = 23.81$$

2. Minimum Hamming Distance

Given codewords:

00000
01011
10101
11110

Calculate the Hamming distance by counting different bit positions.

Codeword pair	**Hamming distance**
00000 and 01011	3
00000 and 10101	3

00000 and 11110	4
01011 and 10101	4
01011 and 11110	3
10101 and 11110	3

Minimum distance:

$$d_{(min)}=3$$

Error detection:

$$d_{(min)}-1=3-1= 2$$

Error correction:

$$d_{(min)} - 1 / 2 = 1$$

Therefore, the code can detect 2 errors and correct 1 error.

3. Hamming (7,4) Error Correction

Given:

- Original codeword = 0110011
- Received codeword = 0110111

The error is at position 5.

Syndrome:

$$s_1 = 1, s_2 = 0, s_3 = 1$$

Error position:

$$pos = 1 + 2(0) + 4(1) = 5$$

Flip bit 5 to correct the error.

Received : 0110111

Corrected: 0110011

Extract data bits from positions 3, 5, 6 and 7.

1011

Result: Corrected data = 1011.

4. Complete the Code – Python Program

```
def encode(d):
    p1 = d[0] ^ d[1] ^ d[3]
    p2 = d[0] ^ d[2] ^ d[3]
    p3 = d[1] ^ d[2] ^ d[3]

    return [p1, p2, d[0], p3, d[1], d[2], d[3]]

def decode(c):
    s1 = c[0] ^ c[2] ^ c[4] ^ c[6]
    s2 = c[1] ^ c[2] ^ c[5] ^ c[6]
    s3 = c[3] ^ c[4] ^ c[5] ^ c[6]

    pos = s1 + 2 * s2 + 4 * s3
```

```
    if pos != 0:
        c[pos - 1] ^= 1

    data = [c[2], c[4], c[5], c[6]]

    return pos, data

word = encode([1, 0, 1, 1])

print("codeword       :", "".join(map(str, word)))

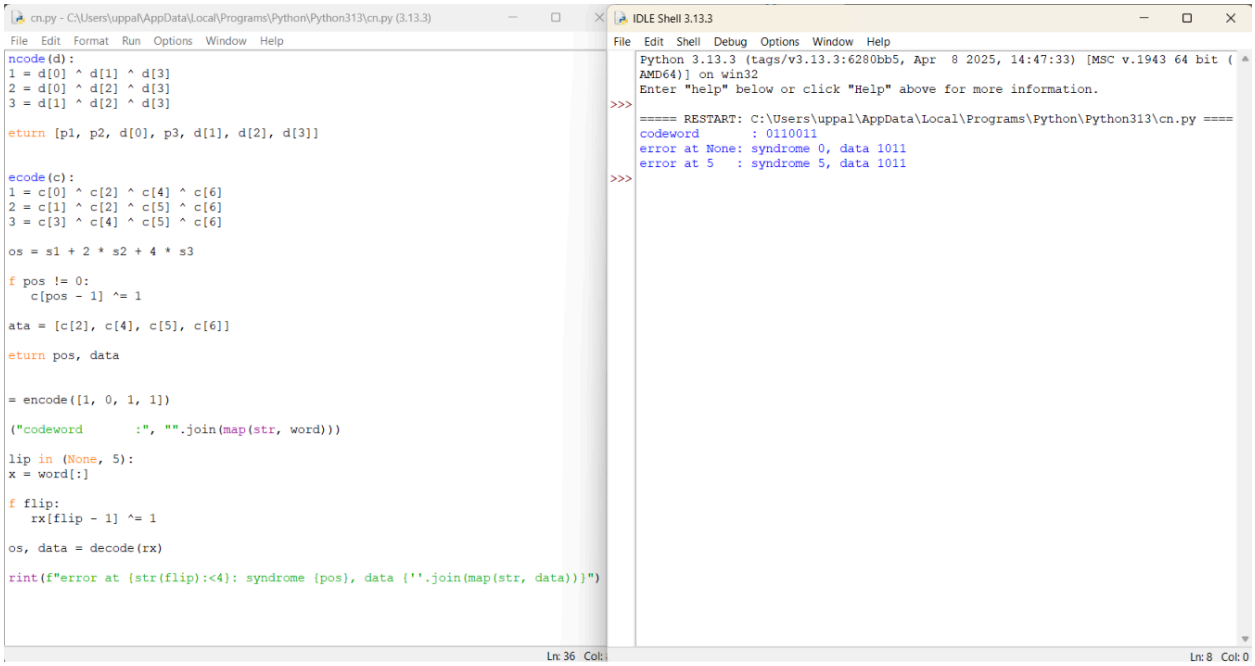
for flip in (None, 5):
    rx = word[:]

    if flip:
        rx[flip - 1] ^= 1

    pos, data = decode(rx)

    print(f"error at {str(flip):<4}: syndrome {pos}, data {''.join(map(str, data))}")
```

Expected Output



5. What Happens When Two Bits Are Flipped?

Original codeword:

0110011

Flip positions 5 and 6:

Received: 0110101

The syndrome becomes 3, so the decoder assumes bit 3 is incorrect and flips it.

The decoder makes another incorrect correction because Hamming (7,4) can correct only single-bit errors.

Therefore, multiple-bit errors can produce incorrect data.

6. Alternative Concept Analysis – FEC vs ARQ

Given:

- Data = 16 bits

- Bit error rate = 1/1000
- Retransmission delay = 40 ms

A. Forward Error Correction (FEC)

Check bits = 5

$$16+5 = 21 \text{ bits}$$

- Adds 5 extra bits.
- Corrects single-bit errors.
- Does not require retransmission for correctable errors.
- Suitable for time-sensitive communication.

B. Automatic Repeat Request (ARQ)

Probability of error in a 16-bit reading:

$$P = 1 - (0.999)^{16}$$

$$P \approx 0.0159$$

Average number of transmissions:

$$\frac{1}{1-P} \approx 1.016$$

Average bits transmitted:

$$16 / 1\text{-}P \sim\sim 16.26 \text{ bits}$$

Average additional retransmission delay:

$$\frac{P}{1-P} \times 40$$

0.645 ms approximately

Comparison

Parameter	**FEC**	**ARQ**
Bits sent	21	16.26 average
Error handling	Corrects errors	Retransmits data
Delay	No retransmission wait	Additional delay
Reliability	Corrects limited errors	Can recover through retries
Suitable for	Real-time communication	Delay-tolerant communication

Why do video streaming and 5G use FEC?

- Reduces the need for retransmission.
- Helps maintain continuous communication.
- Corrects transmission errors.
- Reduces additional delay.

7. Conclusion

Hamming code provides error detection and correction by adding check bits to transmitted data. For a 16-bit word, five check bits are required. FEC is suitable for noisy links where

retransmission takes additional time, while ARQ is useful when retransmission is acceptable. Therefore, FEC is recommended for time-sensitive industrial sensor communication.

TASK 9 – How Big Must the Window Be?

1. Transmission Time and Propagation Time

Given:

- Distance = 400 km = 400,000 m
- Link speed = 1 Gbps = 10^9 bps
- Frame size = 1,500 bytes = 12,000 bits
- Propagation speed = 2×10^8 m/s

A. Transmission Time

$$T_t = \frac{Framesize}{Linkrate}$$

$$= \frac{12000}{10^9}$$

T_t = 12 mu s

B. Propagation Time

$$T_p = \frac{Distance}{Propagationspeed}$$

$$= \frac{400000}{2\times10^8}$$

T_p = 0.002 s = 2 ms

C. Calculate Ratio (a)

$$a = \frac{T_p}{T_t}$$

$$= \frac{0.002}{0.000012}$$

a = 166.67

2. Stop-and-Wait ARQ Utilisation

Formula:

$$U = \frac{1}{1+2a}$$

$$U = \frac{1}{1+2(166.67)}$$

U = 0.299 %

Throughput:

1000 * 0.00299 = 2.99 Mbps

Therefore, Stop-and-Wait uses only a small portion of the available 1 Gbps bandwidth because it waits for an acknowledgement after every frame.

3. Minimum Window Size

For full link utilisation:

$$W \geq 1 + 2a$$

$$W \geq 1 + 2(166.67)$$

$$W \geq 334.33$$

Therefore, the minimum integer window is:

$$W=335 \text{ frames}$$

Sequence Number Bits

Go-Back-N:

$$W \leq 2^k - 1$$

For $k = 9$:

$$2^9 - 1 = 511$$

Therefore, 9 bits are required.

Selective Repeat:

$$W \leq 2^{k-1}$$

For $k = 10$:

$$2^9 = 512$$

Therefore, 10 bits are required.

Protocol	**Minimum sequence bits**
Go-Back-N	9 bits

Selective Repeat	10 bits

4. Go-Back-N vs Selective Repeat

Parameter	**Go-Back-N**	**Selective Repeat**
Error handling	Resends lost frame and subsequent frames	Resends only lost frames
Receiver	Discards out-of-order frames	Stores out-of-order frames
Buffer requirement	Low	High
Efficiency	Lower after frame loss	Higher after frame loss

Evaluation: Selective Repeat resends fewer frames after a loss, but it requires additional receiver memory to store out-of-order frames.

5. HDLC and PPP Frame Delimiters

HDLC and PPP use a special flag sequence to identify the beginning and end of a frame.

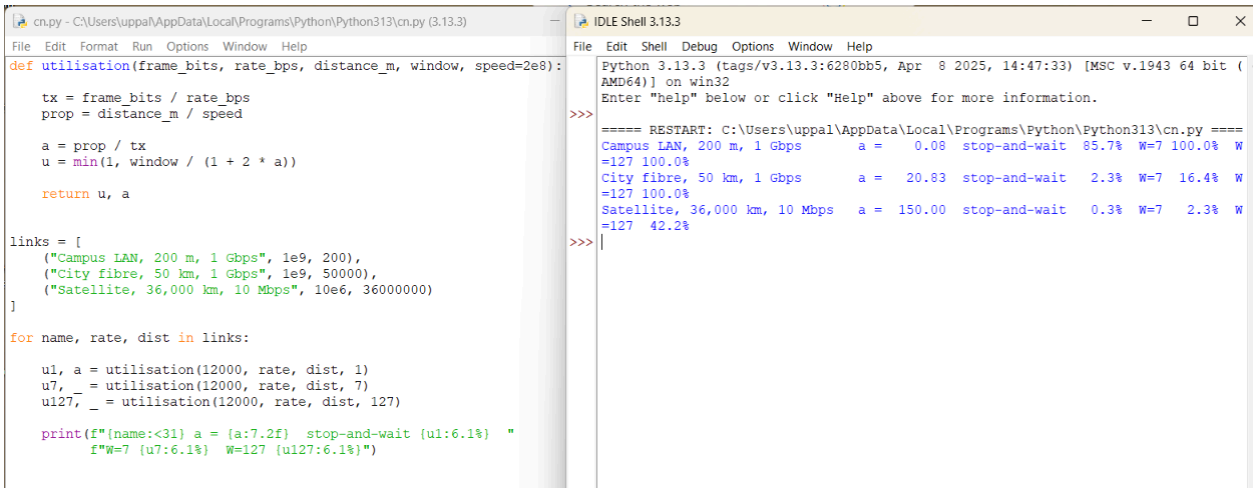
01111110 | DATA | 01111110

- Starting flag identifies the beginning.
- Ending flag identifies the end.
- HDLC uses bit stuffing to prevent data from being mistaken for a flag.
- PPP also uses flag-based framing.

6. Complete the Code – Python Program

```
def utilisation(frame_bits, rate_bps, distance_m, window, speed=2e8):
    tx = frame_bits / rate_bps
    prop = distance_m / speed
    a = prop / tx
    u = min(1, window / (1 + 2 * a))
    return u, a
links = [
    ("Campus LAN, 200 m, 1 Gbps", 1e9, 200),
    ("City fibre, 50 km, 1 Gbps", 1e9, 50000),
    ("Satellite, 36,000 km, 10 Mbps", 10e6, 36000000)
]
for name, rate, dist in links:
    u1, a = utilisation(12000, rate, dist, 1)
    u7, _ = utilisation(12000, rate, dist, 7)
    u127, _ = utilisation(12000, rate, dist, 127)
    print(f"{name:<31} a = {a:7.2f}  stop-and-wait {u1:6.1%}  "
          f"W=7 {u7:6.1%}  W=127 {u127:6.1%}")
```

Output



7. Alternative Concept Analysis – Selective Repeat with Window 64

Given:

- Window size = 64
- Frame size = 1,500 bytes
- Link speed = 1 Gbps
- $a = 166.67$

A. Utilisation

$$U = \frac{64}{1+2(166.67)}$$

$$U = 19.12\ \%$$

B. Throughput

$$1000 \times 0.1912$$

$$191.2\ \text{Mbps}$$

C. Receiver Buffer Space

$$64 \times 1500 = 96000 bytes$$

96,000 bytes = 96 KB

Therefore, a window of 64 requires approximately 96 KB of receiver buffer space but achieves only 19.12% link utilisation.

8. Conclusion

Stop-and-Wait ARQ provides very low utilisation on long-distance links. A window of 335 frames is required to achieve full utilisation on the 400 km fibre link. Selective Repeat is more efficient because it retransmits only lost frames, although it requires more receiver memory. Therefore, Selective Repeat with a sufficiently large window is recommended for reliable and efficient head-office communication.

TASK 10 – Find Where the Delay Comes From

1. Propagation Delay for 400 km Fibre

Given:

- Distance = 400 km = 400,000 m
- Propagation speed = 2×10^8 m/s

Formula:

$$T_p = \frac{Distance}{Speed}$$

$$= \frac{400000}{2 \times 10^8}$$

T_p = 2 ms

This is the one-way propagation delay. The expected round-trip propagation delay is:

$$2 * 2 = 4 \text{ ms}$$

The final traceroute shows approximately 12.6 ms RTT.

Percentage explained by propagation:

$$4 / 12.6 * 100 = 31.75 \%$$

Therefore, fibre propagation explains approximately 31.75% of the measured round-trip delay. The remaining delay comes from routing, processing and queuing.

2. Why Does Hop 3 Show 38.2 ms and Hop 5 Show Stars?

Hop 3 – 38.2 ms

The three measurements are 4.0 ms, 38.2 ms and 4.2 ms. The unusually high value may be caused by temporary congestion or delayed ICMP responses.

It does not necessarily indicate a network fault.

Hop 5 – Stars (*)

Stars indicate that the router did not respond to traceroute probes within the timeout period.

Possible reasons:

- Firewall restrictions.
- ICMP filtering.
- Rate limiting.
- Router security settings.

It does not necessarily mean that the router is faulty.

3. Which Hop Adds the Most Delay?

Hop	RTT observations
1	0.5–0.6 ms
2	2.9–3.4 ms
3	4.0–38.2 ms
4	11.5–12.1 ms
5	No response
6	12.4–12.9 ms

Hop 4 shows the largest stable increase in RTT compared with the normal Hop 3 readings.

Approximately:

$$11.8 - 4.0 = 7.8 \text{ ms}$$

Therefore, the network segment leading to Hop 4 should be investigated for congestion or additional processing delay.

4. Execute Traceroute, Ping and Wireshark

A. Traceroute

Windows:

tracert google.com

Linux:

traceroute google.com

B. Ping

ping google.com

This measures response time and packet loss.

C. Wireshark

1. Open Wireshark.
2. Select your active Wi-Fi or Ethernet interface.
3. Start packet capture.
4. Open a website.
5. Stop the capture.
6. Apply the filter:

icmp || tcp || udp

Identify the headers:

OSI Layer	**Header / Protocol**
Application	HTTP / DNS
Transport	TCP / UDP

Network	IPv4 / IPv6
Data Link	Ethernet / Wi-Fi
Physical	Electrical or radio signals

Evidence requirement: Run these commands on your own computer and attach your actual output screenshots. Do not use the supplied traceroute as your personal execution evidence.

5. Alternative Concept Analysis – Continuous Monitoring

Parameter	Ping and Traceroute	SNMP and NetFlow
Monitoring	Occasional	Continuous or periodic
Packet loss	Detects	SNMP interface counters can detect
Congestion	Limited visibility	SNMP utilization and flow records help identify it
Traffic sources	Limited	NetFlow identifies major traffic sources
Detection speed	During testing	Depends on polling and export intervals

Evaluation: Ping and traceroute are useful for troubleshooting, while SNMP and NetFlow provide continuous visibility into network performance and traffic patterns.

6. Conclusion

The traceroute indicates approximately 12.6 ms round-trip delay, while fibre propagation contributes approximately 4 ms. Hop 3 shows an unusual delay, and Hop 5 does not respond to probes. These observations do not automatically indicate network failures. Continuous monitoring using SNMP and NetFlow, together with ping and traceroute, is recommended for identifying network problems.

TASK 11 – Defend the Network Design

1. Recommended Network Design

The smart manufacturing plant requires a reliable network for robots, sensors, CCTV cameras, office users and head-office communication.

Area	Recommended technology	Reason
Robot cells	Cat6A S/FTP Ethernet	Reduces electromagnetic interference.
Vibration sensors	Industrial Ethernet / Wi-Fi	Supports sensor data transmission.
CCTV cameras	Gigabit Ethernet	Supports continuous video traffic.
Office users	Wi-Fi 6	Provides wireless connectivity.

Shop floor to data centre	Multimode fibre	Supports high-speed communication.
Plant to head office	Single-mode fibre / leased line	Supports long-distance communication.
AGVs	Private 5G	Supports mobility and handovers.

2. Error Control and Flow Control

Network connection	Recommended method	Reason
Robot communication	Ethernet FCS, QoS and TSN	Detects errors and prioritizes time-critical traffic.
Sensor communication	CRC and FEC where appropriate	Helps detect and correct transmission errors.
CCTV communication	Ethernet CRC and TCP where required	Supports reliable video transfer and storage.
Head-office communication	TCP with sliding-window ARQ	Supports reliable long-distance data transfer.

AGV communication	5G reliability mechanisms and redundant communication	Reduces communication interruption.

Supporting calculations:

- Robot traffic = 32 Mbps per production line.
- Total plant traffic = 1265.92 Mbps.
- Head-office fibre propagation delay = 2 ms one-way.
- Stop-and-Wait utilisation = approximately 0.299%.
- A sliding window of 335 frames is required for full theoretical utilisation on the head-office link.

Therefore, sliding-window protocols are preferable to Stop-and-Wait for long-distance communication.

3. Two Biggest Risks

Risk 1: Network congestion

The plant generates approximately 1265.92 Mbps with continuous CCTV streaming. A 1 Gbps uplink is insufficient.

Solution: Use higher-capacity fibre links, QoS and local CCTV recording.

Risk 2: Communication delay

Robot commands require a maximum delay of 10 ms. Network congestion and wireless interference may increase latency.

Solution: Use industrial Ethernet, TSN, private 5G and redundant communication paths.

4. Will the Design Work for Five Plants?

Traffic generated by one plant:

$$1265.92 Mbps$$

Traffic generated by five plants:

$$1265.92 * 5 = 6329.6 \text{ Mbps}$$
$$= 6.33 \text{ Gbps}$$

With local CCTV recording:

$$536.32 * 5 = 2681.6 \text{ Mbps}$$

Therefore, a scalable 10 Gbps head-office backbone is recommended, with appropriate redundancy and capacity planning.

5. Alternative Concept – Entire Plant Using Private 5G

Using Task 1 traffic calculations:

Traffic source	Traffic	Recommendation
Robots	32 Mbps/line	Wired Ethernet for critical control
CCTV	96 Mbps/line	Wired Ethernet
Vibration sensors	0.24 Mbps/line	5G mMTC or industrial wired network
Office users	30 Mbps/line	Wi-Fi 6

Private 5G should primarily support mobile AGVs and suitable wireless sensors rather than replace the entire wired backbone.

Evaluation: A combination of fibre, Ethernet, Wi-Fi 6 and private 5G provides better reliability, flexibility and scalability than using only 5G.

6. Conclusion

The recommended network combines industrial Ethernet, fibre optics, Wi-Fi 6 and private 5G. QoS, TSN and sliding-window communication improve network efficiency. A 10 Gbps backbone can support the estimated five-plant traffic under the local CCTV recording design, subject to sufficient headroom and redundancy. This hybrid architecture provides reliable communication and supports future expansion.

TASK 12 – Re-evaluate and Adapt the Design

A. Impact Analysis – 2 Marks

The company introduces a private 5G network and 50 AGVs to transport materials between production lines.

The changes affect:

- Wireless technology: Private 5G is introduced.
- Robot communication: Requires reliable, low-latency connectivity.
- AGV communication: Requires mobility and fast handovers.
- Error control: Additional wireless error correction and retransmission mechanisms are required.

Calculate Additional AGV Traffic

Assumption: Each AGV generates the same traffic as one robot controller (200 bytes every 2 ms).

Traffic per AGV:

$$\frac{200\times8}{0.002\times10^{6}} = 0.8Mbps$$

For 50 AGVs:

$$50 * 0.8 = 40 \text{ Mbps}$$

Original plant traffic:

$$1265.92Mbps$$

New plant traffic:

$$1265.92+40 = 1305.92 \text{ Mbps}$$

With local CCTV recording:

$$536.32+40 = 576.32 \text{ Mbps}$$

Therefore, the introduction of 50 AGVs increases plant traffic by 40 Mbps under the stated assumption.

B. Test the Original Design – 2 Marks

The original design uses Wi-Fi 6 for wireless connectivity.

Given:

- AGV route = 1200 m
- Speed = 2 m/s
- Wi-Fi access point spacing = 30 m
- Wi-Fi handover = 150 ms
- 5G cell coverage = 300 m

- 5G handover = 20 ms

Wi-Fi Handovers

$$1200/30 = 40 coverage sections$$

$$40 - 1 = 39 handovers$$

Total interruption:

$$39 \times 150 = 5850 ms$$

5.85 seconds

Private 5G Handovers

$$1200/300 = 4 coverage sections$$

$$4 - 1 = 3 handovers$$

Total interruption:

$$3 \times 20 = 60 ms$$

60 ms

Comparison

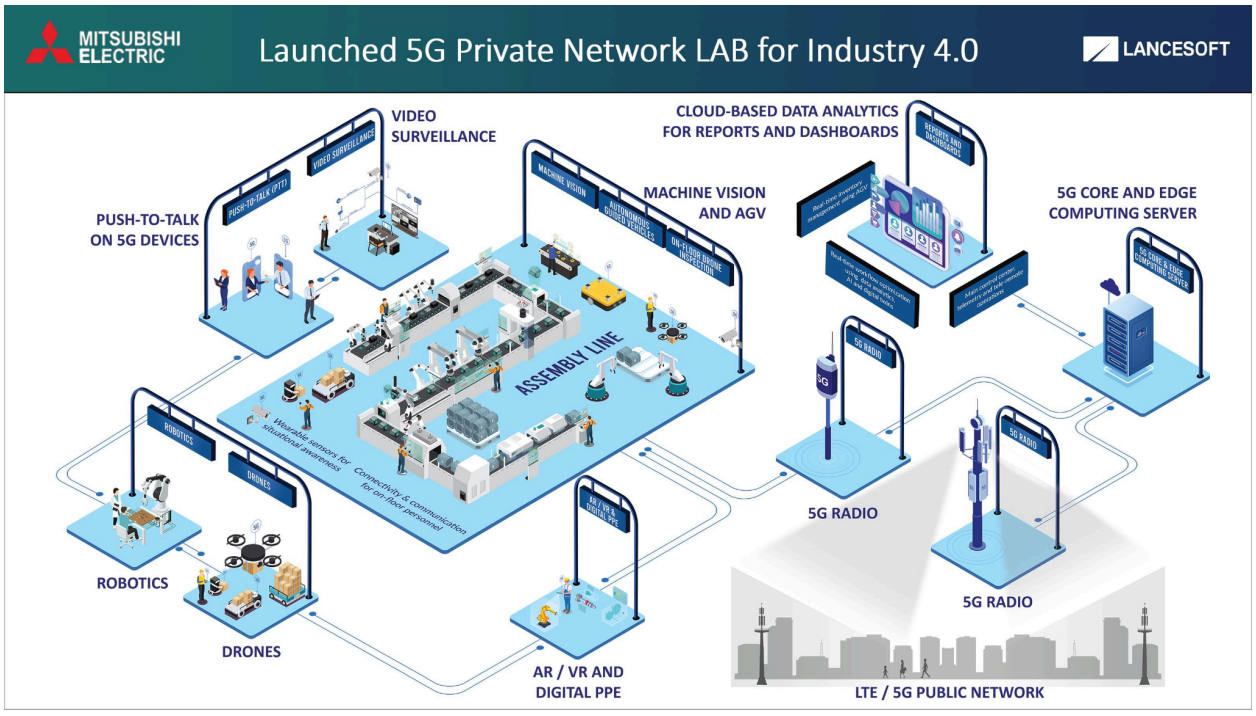
Parameter	Wi-Fi 6	Private 5G
Handovers	39	3

Handover time	150 ms	20 ms
Total interruption	5850 ms	60 ms
10 ms deadline	Not suitable as configured	Not guaranteed

Evaluation: The original Wi-Fi design cannot reliably support the AGVs' 10 ms deadline. Private 5G significantly reduces handover interruptions, but its 20 ms handover time still exceeds the 10 ms requirement.

C. Adaptation or Redesign – 2 Marks

I recommend modifying the original design by introducing a hybrid network architecture.



Proposed Network

Application	Recommended technology
Robot controllers	Industrial Ethernet + TSN
50 AGVs	Private 5G URLLC
CCTV cameras	Gigabit Ethernet
Office users	Wi-Fi 6
Vibration sensors	Industrial Ethernet / suitable wireless
Head-office connection	Single-mode fibre
Plant backbone	10 Gbps fibre

Important improvements:

1. Use private 5G URLLC for AGV communication.
2. Use industrial Ethernet for stationary robot controllers.
3. Implement QoS to prioritize critical commands.
4. Use redundant communication paths.
5. Implement fail-safe stopping when communication is interrupted.
6. Monitor network performance continuously.

D. Evidence-Based Defence – 2 Marks

The modified design is better because:

Parameter	Original design	Modified design
AGV connectivity	Wi-Fi 6	Private 5G
Wi-Fi handovers	39	Not used for AGV control
5G handovers	Not available	3 per trip
AGV traffic	Not included	40 Mbps
Plant traffic	1265.92 Mbps	1305.92 Mbps
Backbone	1 Gbps insufficient	10 Gbps recommended

Evaluation:

- Private 5G reduces the number of handovers from 39 to 3.
- AGV traffic adds 40 Mbps to the plant.
- A 10 Gbps fibre backbone provides sufficient capacity for the estimated traffic.

However, private 5G alone cannot guarantee that commands will never be missed. The system must be tested for latency, packet loss, coverage and handover performance under actual factory conditions.